

Day 105: Weekly Revision — Machine Learning Foundations

1. Day Number

Day 105

2. Topic Name

Revision of Days 99–104: Machine Learning Foundations

3. Connection

Over the last six days, you moved from understanding **what machine learning is** to understanding your first supervised-learning model.

The overall journey was:

```
Dataset
  ↓
Features + Target
  ↓
Train / Validation / Test
  ↓
Watch for Underfitting / Overfitting
  ↓
Understand Bias / Variance
  ↓
Create Useful Features
  ↓
Train a Linear Regression Model
  ↓
Make Predictions
```

This is the beginning of the basic **machine-learning lifecycle**.

4. Revision Summary of Days 99–104

Day 99 — Machine Learning Fundamentals

You learned the main vocabulary:

```
Dataset
Observation
Feature
Label / Target
Model
Prediction
```

An **observation** is usually one row.

A **feature** is information given to the model.

A **target** is what you want the model to predict.

For example:

```
House size
Bedrooms
Age
↓
Features

Price
↓
Target
```

Day 100 — Training, Validation, and Test Data

You learned why we do not normally use the entire dataset for exactly the same purpose.

```
Training set
→ model learns

Validation set
→ helps with development decisions

Test set
→ final evaluation on unseen data
```

The goal is to check **generalization**: whether the model works well on new data.

Day 101 — Underfitting and Overfitting

You learned two common model problems.

Underfitting:

```
Poor training performance
+
Poor test performance
```

The model may not have learned enough.

Overfitting:

```
Excellent training performance
+
Much poorer test performance
```

The model may have learned the training examples too specifically.

Day 102 — Bias and Variance

You connected bias and variance to the previous lesson.

```
High bias
→ model may be too simple
→ underfitting
```

```
High variance
→ model may depend too much on training details
→ overfitting
```

The goal is good **generalization**, rather than simply creating the most complicated model possible.

Day 103 — Feature Engineering

You learned that useful features can sometimes be created from raw data.

For example:

```
transaction_date
  ↓
transaction_month
```

or:

```
customer_age
  ↓
age_group
```

A derived feature should:

- make sense for the prediction problem
 - contain useful information
 - be available when the prediction is actually made
-

Day 104 — Linear Regression

You trained toward your first simple supervised regression model.

Linear Regression learns a relationship similar to:

```
prediction =
coefficient × feature + intercept
```

For example:

```
House Size
  ↓
Linear Regression
  ↓
Predicted House Price
```

You also learned the basic scikit-learn workflow:

```
prepare X and y
  ↓
create model
  ↓
fit()
  ↓
predict()
```

5. Important Topics

Here are the main concepts to remember from this week.

Concept	Beginner Meaning
Machine Learning	Learning patterns from data
Feature	Input information
Label / Target	Value we want to predict
Training Set	Data used to learn
Validation Set	Data used for development decisions
Test Set	Data reserved for final evaluation
Underfitting	Model has not learned enough
Overfitting	Model learned training data too specifically
High Bias	Often associated with an overly simple model
High Variance	Often associated with sensitivity to training data
Feature Engineering	Creating useful features from existing data
Linear Regression	Predicting numerical targets using a linear relationship

6. Foundational Notes

A machine-learning system does not begin with the model.

You normally begin with the **problem and data**.

Ask:

What am I trying to predict?

↓

Target

What information is available?

↓

Features

What examples can the model learn from?

↓

Training data

How will I check generalization?

↓

Validation / test data

Only then should you think about training a model.

Also remember:

┆ A high training score does not automatically mean you have a good model.

A useful model should perform reasonably on **unseen data**.

7. Easy Example

Suppose you have:

Size	Bedrooms	Age	Built Year	Price
900	2	15	2011	45
1200	2	10	2016	58
1500	3	7	2019	72
1800	3	4	2022	88
2100	4	2	2024	105

Suppose your goal is:

┆ Predict the selling price of a house.

You should immediately begin asking:

What is the target?

Which columns could be useful features?

Could any useful feature be derived?

How should the observations be divided?

What could cause overfitting?

Is Linear Regression appropriate as a simple first model?

Those questions combine almost everything from Days 99–104.

8. Revision Problem Statement

Consider this tiny house dataset:

House	Size	Bedrooms	Built Year	Price
A	1000	2	2012	50
B	1300	2	2016	64
C	1600	3	2019	78
D	1900	3	2021	91
E	2200	4	2024	108
F	2500	4	2025	120

Your task is to explain:

1. Which column is the **target** if you want to predict house price?
2. Which columns could be used as **features**?
3. How could these observations conceptually be separated into training, validation, and test groups?
4. What is one possible **overfitting risk**?
5. What simple new feature could you derive from Built Year?
6. How would **Linear Regression** conceptually learn from the data?
7. What information would you provide when asking the trained model to make a new prediction?

Do not build a large project.

9. Concepts Used

The revision problem combines:

observations
features
target
supervised learning
regression
training data
validation data
test data
generalization
underfitting
overfitting
bias
variance
feature engineering
Linear Regression
prediction

The important skill is seeing how these ideas fit together rather than treating each one as an isolated definition.

10. Thought Process

Start with the prediction objective.

Ask:

| What exactly do I want to predict?

That determines your **target**.

Then ask:

| What information would be available before making that prediction?

Those columns are possible **features**.

Next think about model development:

Available dataset
↓
Training portion
Validation portion
Test portion

Then consider model behavior:

Does it learn too little?

→ possible underfitting / high bias

Does it learn the training data too specifically?

→ possible overfitting / high variance

Then look for meaningful transformations:

Built Year

↓

Could another useful feature be calculated from it?

Finally think about Linear Regression:

Features

↓

fit()

↓

Learn relationship with price

↓

New house features

↓

predict()

11. Beginner-Friendly Pseudocode

```
START

load or create house dataset

identify target:
    what are we predicting?

identify useful features:
    what information helps make the prediction?

create any sensible derived feature

separate data conceptually into:
    training
    validation
    test

prepare:
    X = features
    y = target

create Linear Regression model

train model using training data

use validation data while making development decisions

when model development is finished:
    evaluate using test data

give model features from a new house

make prediction

END
```

For today's revision, you do not need to implement every step.

12. Suggested Solving Approach: Conceptual + scikit-learn Workflow

Solve the exercise in two parts.

Part A — Conceptual

Identify:

```
observation
features
target
derived feature
data split
possible overfitting risk
```

Make sure you can explain **why** each choice makes sense.

Part B — scikit-learn Workflow

Think about the future Python structure:

```
X = selected features
y = target

create LinearRegression model

model.fit(...)

model.predict(...)
```

Do not worry about advanced tuning yet.

The important distinction is:

```
fit()
→ learning

predict()
→ using what was learned
```

13. Easy Edge Cases

Very Small Dataset

The example contains only six houses.

Dividing six observations into three groups leaves very little data in each group.

For learning purposes, you can still discuss the split conceptually, but a real ML model would usually need considerably more useful data.

Missing Feature

Suppose a new house has:

```
Size = 1800
Bedrooms = missing
```

If bedrooms are required by the trained model, you must decide how to handle that missing information.

New Value Outside Training Range

Suppose the largest training house was:

2500 sq ft

but you ask for a prediction for:

10000 sq ft

Linear Regression can produce a number, but the prediction may be unreliable because this is far outside the training range.

Useless Derived Feature

Creating more columns does not automatically improve a model.

For example:

Size_Copy = Size

provides essentially the same information as Size.

Missing Target

If some training houses have no known price, they cannot straightforwardly provide the target information needed for ordinary supervised training.

14. Common Mistakes to Avoid

- Confusing a **feature** with the **target**.
 - Using the test set repeatedly to decide how to change the model.
 - Assuming the highest training performance means the best model.
 - Thinking every complex model is automatically better.
 - Assuming every new derived feature improves performance.
 - Creating features using information that would not exist at prediction time.
 - Forgetting that X contains input features while y contains the target.
 - Confusing `fit()` with `predict()`.
 - Assuming correlation learned by Linear Regression automatically proves causation.
 - Trusting predictions far outside the range of the training data without caution.
-

15. Quick Self-Check Questions

1. If you want to predict house price, is Price normally a **feature** or a **target**?

2. Which dataset should generally remain separate until final model evaluation: **training**, **validation**, or **test**?
3. A model gets excellent training performance but much poorer test performance. Which concept should you investigate: **underfitting** or **overfitting**?
4. Which is more closely associated with underfitting: **high bias** or **high variance**?
5. In scikit-learn, what is the basic difference between `fit()` and `predict()`?

Try answering these without checking the earlier sections.

16. Hint Only

For the revision problem, begin with the sentence:

| I want to predict __ using information about __.

That should help you separate the target from the features.

For `Built Year`, think about information that could describe **how old a house is**.

For the model-development process, remember:

Training → learn
Validation → guide development
Test → final check

And when thinking about Linear Regression:

Known house features + known prices
↓
<code>fit()</code>
↓
learned relationship
↓
new house features
↓
<code>predict()</code>

For **Day 105**, focus on connecting the pieces. You now have the basic path from **raw data** → **features and target** → **data splitting** → **generalization** → **feature engineering** → **first regression model**.